Wrik Bhadra

email: wrik-bhadra@uiowa.edu / phone: +1-(319)-333-9843
address: 5612 Seamans Center, The University of Iowa, Iowa City, Iowa 52242, USA
website: https://mavewrik.github.io

Research Interests

Medical Image Analysis, Computer Vision, Machine Learning

Work Experience

The University of Iowa, Iowa City, USA

- *Graduate Research Assistant* — Jan 2023 – present
  Roy J. Carver Department of Biomedical Engineering

Rakuten India, Bangalore, India

- *Software Engineer II - Machine Learning* — Jan 2022 – Oct 2022
- *Software Engineer I - Machine Learning* — Aug 2021 – Dec 2021
- *Software Engineer I* — Aug 2020 – Jul 2021

Education

The University of Iowa, Iowa City, USA
Graduate student in Biomedical Engineering — Jan 2023 – present

IIIT Delhi, New Delhi, India
M.Tech. in Computer Science and Engineering — Aug 2018 – Aug 2020

Publications

Transcriptional advantage influence odorant receptor gene choice
Briefings in Functional Genomics (Oxford Academic journal) [published Dec 2022]
https://doi.org/10.1093/bfgp/elac052

Method, Apparatus, and Computer Readable Medium (Patent)
US Patent and Trademark Office (app. number: 17/645,726) [filed Dec 2021]

Generalized Prediction of Hemodynamic Shock in Intensive Care Units
AAAS Science Translational Medicine
medRxiv 2021.01.07.21249121

Certificates

- Graduate Certificate in Biostatistics, The University of Iowa — May 2026

Conferences

- Rising BME Scholars Regional Conference 2024, University of Illinois Chicago — Jun 2024
  Poster: Shape priors improve lung segmentation in ARDS-afflicted subjects

Invited Talks

- ECE:5995 - Applied Machine Learning, The University of Iowa — Feb 2024
  Topic: Lung nodule detection and classification for cancer diagnosis
  Host: Dr. Mathews Jacob

Selected Projects

Automated BMO segmentation in swollen optic nerve head OCT

{*segmentation, multi-task learning, shape priors, causal inference*}
Deep learning framework for automated segmentation of Bruch's membrane opening (BMO) in OCT of the optic nerve head with papilledema, using longitudinal clinical data.

- Built a multi-task deep learning pipeline for BMO segmentation and center localization in swollen optic nerve head OCT scans, where edema obscures anatomical boundaries.
- Developed a denoising signed distance map variational autoencoder (SDM-VAE) shape prior to refine segmentation predictions and encourage anatomically-plausible BMO contours.
- Applied mediation analysis to investigate whether OCT shadowing mediates the relationship between papilledema severity and segmentation performance.
- Designed counterfactual interventional experiments by introducing controlled synthetic shadowing and used repeated measures mixed-effects models to estimate changes in segmentation accuracy, while holding underlying anatomy fixed.

Shape priors for injured lung segmentation

{*segmentation, shape priors, large-scale training*}
Deep learning based automated lung segmentation framework for animal models of ARDS injury (Acute Respiratory Distress Syndrome).

- Developed project from ground-up to incorporate shape information using VAEs for better injured lung segmentation in multi-center multi-species animal dataset (porcine, ovine, canine).
- Developed the framework to train and analyze several deep learning models with extensive hyperparameter tuning on SGE High Performance Computing (HPC) systems.
- Incorporated batch processing to cut down inference time drastically (from about 12 hrs to less than 20 mins overall, for 100 test images each taking roughly 8 mins sequentially) .
- Implemented solutions to generate and visualize experiment intermediates during model training using Tensorboard in PyTorch.
- Additionally, performed statistical analysis using Python packages to compare and visualize performance of various baseline and proposed methods on test dataset.

Relevant Coursework

- Causal Inference - The University of Iowa (Spring 2026)
- Principles of MRI - The University of Iowa (Spring 2026)
- Longitudinal Data Analysis - The University of Iowa (Fall 2025)
- Applied Survival Analysis - The University of Iowa (Spring 2025)
- Deep Learning in Medical Imaging - The University of Iowa (Fall 2024)
- Regression & ANOVA in Health Sciences - The University of Iowa (Spring 2024)
- Medical Imaging Physics - The University of Iowa (Fall 2023)
- Applied Machine Learning - The University of Iowa (Spring 2023)

Skills

Machine Learning frameworks PyTorch, scikit-learn
Packages / Softwares 3DSlicer, MONAI, SimpleITK, TensorBoard, pandas, numpy, matplotlib
Programming languages Python, MATLAB
Application development Flask, MongoDB, SQL, Docker, Kafka
Tools Bash, Git, LaTeX, Zotero, Jupyter Notebook/Lab, Agentic coding
Statistics Causal inference, Survival analysis, Longitudinal data analysis

Service

The University of Iowa

- Summer 2024 – Mentored a BME undergrad student on a project involving lung segmentation on human 3DCT scans.
- Apr 2024 – Invited by the BME Department to moderate a session for BME Senior Design Day 2024.
- Mar 2024 & Summer 2023 – Conducted initial discussion-cum-interview for five prospective graduate students.
- Spring 2024 & Fall 2023 – Peer Mentor for two incoming international graduate students under the iPeer Mentorship Program.

Rakuten India

- Jun 2021 & Mar 2021 – Conducted technical interviews for a prospective full-time and three intern candidates for a separate business unit.
- Nov 2020 - Jan 2021 – Mentored a CS undergrad intern on a project involving application of image processing techniques.
- Feb 2020 – Attended and manned booth on behalf of Rakuten India as tech representative. certificate